Moulik Kumar

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Summary

M.S. Data Science candidate at the University of Colorado Boulder with experience building analytics pipelines and production machine learning systems. Proficient in SQL and Python (Pandas, NumPy, scikit-learn) for large-scale data analysis, reporting, and predictive modeling. At PubMatic, delivered data-driven workflows supporting 200+ advertisers and contributing to $18M+ in programmatic revenue impact. Background in quantum computing and scientific machine learning, with strong ability to translate complex technical data into actionable insights for cross-functional stakeholders.

Education

University of Colorado, Boulder Aug 2025 – Jun 2027
Master of Science, Data Science

Indian Institute of Technology, Kharagpur Aug 2021 – May 2025
Bachelor of Science, Physics

Technical Skills

Programming Languages & Tools: Python, SQL, R, Git, Jupyter Notebook
Data Analysis & Visualization: NumPy, Pandas, SciPy, scikit-learn, matplotlib, Seaborn
Machine Learning: Linear Regression, Decision Trees, Random Forests, Gradient Boosting (XGBoost), Neural Networks, Deep Learning (TensorFlow, PyTorch), NLP
Mathematics & Statistics: Linear Algebra, Probability Theory, Statistical Modeling, Numerical Analysis
Domain Experience: Computational Physics, Quantum Computing (Qiskit), Graph Neural Networks

Professional Experience

PubMatic Inc May 2024 – Aug 2024
CTO Team Intern Pune, Maharashtra

- Designed and deployed multimodal models (audio, OCR, vision) with scalable inference pipelines for low-latency ad classification and decision signals supporting ad delivery and monetization workflows.
- Queried and analyzed large-scale advertiser datasets using SQL and Python (Pandas, NumPy) to generate interpretable insights on issue detection and sentiment trends.
- Developed and validated candidate verification pipelines using fuzzy matching and NER techniques, achieving 98.7% precision in identifying political figures across high-volume creative datasets.
- Contributed to the launch of AdWise, a production ML system supporting 200+ advertisers and driving $18M+ incremental programmatic revenue during the 2024 election cycle.

Indian Institute of Science Education and Research (IISER), Bhopal Jul 2024 – Sep 2024
Summer Research Intern Remote

- Designed and implemented a hybrid deep learning model (BiLSTM + Transformer-based attention) integrating TF-IDF, semantic, and syntactic features to classify subgenres within the Brown corpus.
- Conducted extensive experimentation with linguistic features, n-gram TF-IDF, and character-level encodings, achieving measurable improvements in F1 scores.
- Performed systematic hyperparameter tuning, regularization, and data augmentation to optimize model performance, leveraging analytical skills to analyze results and creating data visualizations to communicate improvements and ensure reproducible, data-driven outcomes.

Publications and Conferences

- Multi-Modal Machine Learning for Political Video Advertisement Analysis: Integrating Audio, Textual, and Visual Features – International Journal of Computer Applications. 186, 46 (November 2024), 49–55. DOI=10.5120/ijca2024924115
- Building Resilience through ESG: Evaluating Indian PSUs in the Context of Global Emerging Markets – 11th International Conference on Business Analytics and Intelligence, Indian Institute of Management Bangalore (December 2024).

Major Projects

Machine Learning–Driven Discovery of Stable Heusler Alloys
Undergraduate Thesis

- Built an end-to-end ML pipeline to predict thermodynamic stability of 1000+ Heusler alloys using engineered compositional and structural features.
- Trained and optimized an XGBoost classifier (stratified split, cross-validation) achieving 99% test accuracy; applied SHAP and Partial Dependence analysis to interpret key stability drivers.
- Implemented Crystal Graph Convolutional Neural Networks (CGCNN) to predict formation energy (R^2 = 0.92, MAE = 0.035 eV/atom), modeling atomistic interactions via graph-based embeddings.
- Developed a Conditional WGAN-GP to generate synthetic alloy candidates conditioned on magnetic targets; screened 500 generated samples with 65% predicted stable.

Building Resilience through ESG: Evaluating Indian PSUs in the Context of Global Emerging Markets

- Developed econometric models (static panel regression, GMM estimators) and stress-tested 33 Indian PSUs (-10%, -20%, -30% shocks), showing a 3.3 basis point ROA gain per unit ESG score.
- Built ESG performance metrics integrating financial indicators and NLP-based ML models, improving governance resilience predictions by 15%.
- Conducted comparative ESG analysis across emerging markets, revealing a 38.9% performance drop for low-ESG PSUs under severe stress, with actionable sustainability insights.

Simlulating the Prisoner's Dilemma in a Quantum Environment

- Modeled the Quantum Prisoner's Dilemma using IBM Qiskit, implementing parameterized unitary strategy operators and entangling circuits.
- Simulated quantum strategies under varying entanglement parameters and computed expected payoffs using probabilistic measurement outcomes.
- Demonstrated convergence of Nash and Pareto equilibria under maximal entanglement, validating theoretical results from quantum game theory literature.

Scholastic Achievements

- Secured All India Rank of 6,767 in JEE Advanced 2021 (out of 172,000 candidates).
- Top 1.5% all over India in JEE Mains 2021 (out of 1,200,000 candidates).
- Qualified for INSPIRE Scholarship, issued by the Department of Science and Technology, Government of India.